

# Optimal Depth of Multimodal Fusion and Mitigating Modality Collapse

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**Abstract.** Multimodal Machine Translation (MMT) aims to resolve linguistic ambiguities by incorporating visual context. However, models frequently suffer from modality collapse, where the decoder relies entirely on strong textual priors and completely ignores the visual stream. In this paper, we investigate the root causes of this phenomenon using our previously developed Dual Attention Decoder (DAD) baseline. To enforce visual grounding, we propose three interconnected methodologies: (1) a structural search for optimal cross-modal fusion depth, (2) Text-Guided Attention Pooling to dynamically extract relevant visual patches, and (3) an InfoNCE auxiliary loss to synchronize text and image representations in the latent space. Extensive evaluations on the Multi30k and MSCOCO datasets reveal a critical architectural paradox. While top-heavy fusion with InfoNCE alignment achieves the highest general translation fluency (41.69 BLEU), performance on the CoMMuTE visual disambiguation benchmark remains strictly stagnant at approximately 50.00%—statistical random chance. Our failure analysis demonstrates that encoder-level semantic alignment is fundamentally insufficient because the decoder’s static fusion mechanism mathematically incentivizes ignoring visual features to avoid noise. We conclude that future MMT architectures must move beyond static structural routing and integrate dynamic, bias-mitigating mathematical penalties to enforce true multimodal dependency.

**Keywords:** Multimodal Machine Translation · Modality Collapse · Vision Transformer · Information Bottleneck · Contrastive Learning.

## 1 Introduction

Standard Neural Machine Translation (NMT) has achieved remarkable success by leveraging massive text corpora and advanced transformer architectures. However, text-only models inherently suffer from a critical limitation: they cannot resolve linguistic ambiguities—such as gender, color, and referential polysamy—that require external context. Multimodal Machine Translation (MMT) introduces visual features alongside textual inputs to bridge this gap, aiming to utilize situational imagery as a grounding mechanism to disambiguate target sentence generation.

Despite the theoretical promise of MMT, practical implementations frequently encounter a foundational obstacle known as "Modality Collapse". Because language models are trained on highly structured textual statistics, the decoder often relies entirely on strong language priors, completely ignoring the visual stream during training. This phenomenon represents the path of least mathematical resistance, where the model minimizes loss by skipping the complex cross-modal alignment required to interpret images. Compounding this issue is an evaluation gap: standard automated metrics, such as BLEU, primarily measure surface-level linguistic fluency rather than true cross-modal reasoning. Consequently, a model can achieve competitive BLEU scores while remaining fundamentally blind to the visual context, highlighting the strict necessity for targeted visual grounding benchmarks such as CoMMuTE [1].

To address the challenges of effective cross-modal integration, we previously developed the Dual Attention Decoder (DAD) architecture. DAD establishes independent cross-attention mechanisms for text and vision within the decoder stack, combining them via a static 0.5\_sum fusion scheme. While this baseline successfully delivers highly robust and stable general translation quality across multiple domains, our subsequent rigorous evaluations on the CoMMuTE benchmark revealed a persistent vulnerability: the model still succumbs to complete modality collapse. Because the static fusion weights force an equal split regardless of contextual difficulty, the decoder optimizes for safety by nullifying the visual weights during straightforward textual sequences, ultimately failing to "see" the image even when it is explicitly required.

In this paper, we systematically investigate the architectural and semantic root causes of this failure. First, we explore the structural question of fusion depth. We hypothesize that as representations propagate through the decoder, higher layers process increasingly abstracted and semantic content. Therefore, we theorize that the model may not require visual features at early syntactic stages, but rather needs them at these later, highly abstracted stages to resolve deep semantic ambiguities. To validate this, we implement and compare bottom-heavy, top-heavy, and all-layer cross-attention configurations. Building upon the relatively superior performance of the top-heavy approach, we further expand our structural search by meticulously fine-tuning the ratio between text-only layers and multimodal fusion layers. Second, to ensure high-fidelity visual delivery, we introduce Text-Guided Attention Pooling. Instead of naive mean pooling or static CLS tokens that destroy fine-grained visual details, our mechanism utilizes the global text vector as a dynamic query to selectively extract semantically relevant patches from a Vision Transformer (ViT) [2]. Third, we implement an encoder-level InfoNCE auxiliary loss to mathematically synchronize the visual and textual semantic spaces prior to decoding.

Our extensive empirical evaluations on the Multi30k and MSCOCO datasets yield critical, cautionary insights for the MMT community. We demonstrate that altering the architectural depth of fusion is a structural illusion; despite our abstraction hypothesis, CoMMuTE scores remain firmly stagnant at approximately 50.00%—the statistical equivalent of random guessing—regardless

of where cross-attention occurs. Furthermore, while pairing top-heavy fusion with InfoNCE alignment achieves the highest general translation fluency (41.69 BLEU), encoder-level synchronization alone is fundamentally insufficient to break the decoder’s text bias. These findings expose the inherent limitations of rigid, static fusion mechanisms and establish the critical necessity for future dynamic, decoder-level mathematical penalties to enforce true multimodal dependency.

Background and Baseline

### 1.1 The Dual Attention Decoder (DAD) Architecture

To establish a foundation for our investigation, we detail the architecture of the Dual Attention Decoder (DAD), which serves as our strong baseline. We initially engineered the DAD architecture to effectively process textual and visual modalities in parallel. Unlike standard NMT decoders that attend solely to the source text, DAD employs independent cross-attention mechanisms for both the text and image streams within each decoder layer. To merge these parallel contextual representations into a single unified vector for target word generation, the model utilizes a static fusion mechanism, mathematically defined as:

$$Output_t = 0.5 \times \text{Text} + 0.5 \times \text{Image} \quad (1)$$

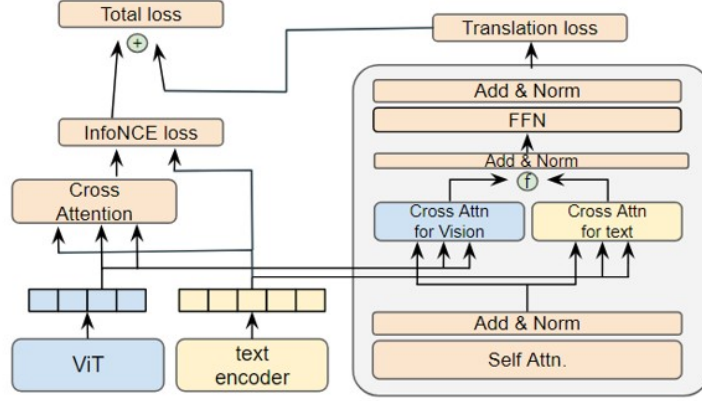
During the preliminary development of this architecture, dynamic gating mechanisms were extensively tested. However, empirical evaluations consistently demonstrated that the fixed 0.5\_sum strategy proved the most robust for maintaining general translation fluency across various hyperparameters, establishing it as our primary baseline.

### 1.2 Visual Feature Extraction via Vision Transformer

To process the visual input, our baseline employs a Vision Transformer (ViT) [2]. Unlike traditional Convolutional Neural Networks (CNNs) that produce spatial feature maps, ViT divides an image into a sequence of fixed-size patches (e.g., 16x16 pixels). These patches are linearly projected into 196 flattened sequence tokens. While this tokenized representation is highly compatible with the transformer-based DAD architecture, it introduces an inherent structural challenge: the network must somehow reconcile a single textual semantic context with 196 separate, uncompressed visual patches.

### 1.3 The Trap of Static Fusion and Modality Collapse

While the DAD architecture yields highly competitive BLEU scores, its reliance on a static fusion mechanism introduces a critical vulnerability. The 0.5\_sum equation rigidly forces the decoder to allocate equal weight to both modalities at every decoding step. Consequently, when the model encounters straightforward textual sequences, the mandatory 50% visual input introduces potential informational noise. To minimize translation loss, the decoder optimizes for safety



**Fig. 1.** Overall architecture of the proposed multimodal translation model integrating Text-Guided Attention Pooling and InfoNCE auxiliary loss.

by driving the visual attention weights toward zero, effectively shutting down the visual stream. This learned blindness—ignoring the 196 visual patches to rely purely on text—is the direct catalyst for modality collapse. Recognizing this structural trap necessitates our exploration into architectural depth and semantic alignment to force the decoder to break this text-only bias.

## 2 Proposed Methods

To systematically mitigate modality collapse and enforce meaningful visual utilization, we propose three interconnected methodologies. As illustrated in Fig. 1, the overall architecture integrates: (1) an architectural exploration of optimal fusion depth alongside hybrid layer ratios, (2) a text-guided attention pooling mechanism to preserve fine-grained visual features, and (3) an encoder-level contrastive alignment bridge utilizing InfoNCE loss.

### 2.1 Searching for Optimal Fusion Depth and Hybrid Layer Ratios

We first investigate the structural hypothesis that the positioning of cross-modal fusion within the decoder dictates the model’s susceptibility to modality collapse. We theorize that as representations propagate through the transformer stack, lower decoder layers primarily resolve syntactic and surface-level textual structures, whereas higher decoder layers process highly abstracted semantic content. Consequently, the explicit dependency on visual context may only become critical at later stages of decoding where deep semantic ambiguities must be resolved.

To rigorously evaluate this, we systematically alter the boundary where the decoder begins to attend to the visual stream, implementing three primary configurations:

- **Bottom-heavy Fusion:** Cross-modal attention is restricted to the initial layers of the decoder to ground the model early in the visual space.
- **Top-heavy Fusion:** Cross-modal attention is reserved strictly for the later layers of the decoder, preserving language fluency in the early stages.
- **All-layers Fusion (Baseline DAD):** Cross-modal attention is uniformly applied across every layer of the decoder stack.

Building upon the initial empirical finding that the top-heavy approach yields relatively superior translation fluency, we expand our structural search to explore the fine-grained boundary between text-only self-attention and multimodal cross-attention. Specifically, we design two hybrid architecture variants to evaluate the optimal macro-demarcation of modalities:

- **text1dual3:** The decoder consists of 1 text-only self-attention layer followed by 3 parallel dual-attention (DAD) fusion layers.
- **text3dual1:** The decoder consists of 3 text-only self-attention layers followed by 1 final parallel dual-attention (DAD) fusion layer.

By evaluating these variants, we aim to isolate whether the sheer volume of multimodal layers or the specific entry point of visual information governs the retention of visual semantics.

## 2.2 Text-Guided Attention Pooling

As established in Section 2.2, a fundamental mismatch exists between the single textual vector representing sentence-level context and the 196 flattened spatial tokens emitted by the Vision Transformer (ViT). Naive compression strategies, such as global average pooling, inherently smooth out localized features, causing critical fine-grained details (e.g., a small distinct object) to blend catastrophically into the background noise.

To preserve spatial fidelity while producing a unified cross-modal bridge, we introduce Text-Guided Attention Pooling. Instead of compressing the image patches blindly, we utilize the global textual representation as a dynamic query to actively filter and extract semantically relevant visual regions.

Let  $\mathbf{h}_{text} \in \mathbb{R}^d$  represent the global context vector derived from the text encoder, and let  $\mathbf{V}_{vit} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{196}\} \in \mathbb{R}^{196 \times d}$  denote the sequence of visual patch tokens generated by the ViT. The text-guided pooling operation is mathematically formulated as a cross-attention mechanism:

$$\mathbf{Q} = \mathbf{h}_{text} \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{V}_{vit} \mathbf{W}_K, \quad \mathbf{V} = \mathbf{V}_{vit} \mathbf{W}_V \quad (2)$$

$$\alpha_i = \frac{\exp\left(\frac{\mathbf{Q} \cdot \mathbf{k}_i^T}{\sqrt{d}}\right)}{\sum_{j=1}^{196} \exp\left(\frac{\mathbf{Q} \cdot \mathbf{k}_j^T}{\sqrt{d}}\right)} \quad (3)$$

$$\mathbf{z}_{visual} = \sum_{i=1}^{196} \alpha_i \mathbf{v}_i \quad (4)$$

where  $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d}$  are learnable projection matrices,  $\alpha_i$  represents the contextual relevance score of the  $i$ -th visual patch relative to the source text, and  $\mathbf{z}_{visual} \in \mathbb{R}^d$  is the final, highly refined single global visual vector delivered to the decoder. This mechanism ensures that the visual stream selectively delivers only the imagery directly corresponding to the linguistic focus of the translation task.

### 2.3 Cross-Modal InfoNCE Alignment

Providing high-fidelity visual features via attention pooling is insufficient if the decoder remains biased toward text-only generation. To proactively bridge the semantic gap before the features enter the decoder stack, we implement an encoder-level auxiliary alignment objective utilizing the InfoNCE (Information Noise-Contrastive Estimation) [3] loss function.

The InfoNCE loss acts as a cross-modal dictionary synchronization mechanism. Given a batch of  $B$  paired instances, let  $\mathbf{h}_{text}^b$  and  $\mathbf{z}_{visual}^b$  represent the synchronized textual and pooled visual vectors for the  $b$ -th sample, respectively. The contrastive loss for a single batch is defined as:

$$\mathcal{L}_{\text{infoNCE}} = -\frac{1}{B} \sum_{b=1}^B \log \frac{\exp(\text{sim}(\mathbf{h}_{text}^b, \mathbf{z}_{visual}^b)/\tau)}{\sum_{k=1}^B \exp(\text{sim}(\mathbf{h}_{text}^b, \mathbf{z}_{visual}^k)/\tau)} \quad (5)$$

where  $\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}^T}{\|\mathbf{u}\| \|\mathbf{v}\|}$  denotes cosine similarity, and  $\tau$  is a temperature hyperparameter controlling the scale of penalties for negative pairs. By maximizing the mutual information between matching text-image pairs while penalizing mismatched pairs within the mini-batch, the encoder forces the visual features into the identical latent space occupied by the source text language.

Ultimately, the model is optimized in a multi-task learning framework. The global objective function combines the standard cross-entropy translation loss ( $\mathcal{L}_{\text{trans}}$ ) computed at the end of the decoder with our auxiliary alignment loss, expressed via direct joint addition:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{trans}} + \mathcal{L}_{\text{infoNCE}} \quad (6)$$

This joint optimization mathematically guarantees that the network must simultaneously learn to generate fluent translations and maintain cross-modal semantic cohesion.

## 3 Experimental Setup

### 3.1 Datasets

To evaluate the translation fluency and generalization capabilities of our proposed multimodal architectures, we utilize two established benchmarks. Our primary in-domain training and evaluation are conducted on the Multi30k dataset [5],

specifically focusing on the English-to-German (En-De) translation task. To further assess the models’ robustness and out-of-domain generalization, we perform cross-dataset evaluations on the MSCOCO dataset [6], which features a broader and more complex vocabulary distinct from the Multi30k training distribution.

### 3.2 Implementation Details and Vision Backbone

The text encoder and the dual attention decoder are implemented based on a standard Transformer architecture configuration [4], but are appropriately scaled for our specific tasks. While the original Transformer architecture [4] conventionally stacks  $N=6$  layers, we configure both our text encoder and the dual attention decoder with  $N=4$  layers. This deliberate structural reduction strictly follows the validated DAD baseline architecture. It not only optimizes parameter efficiency but also acts as a natural regularization mechanism to mitigate the risk of overfitting on the relatively small-scale Multi30k dataset. For the visual feature extraction, we strictly utilize the ViT-Tiny (16x16 patch size) pre-trained backbone.

During our preliminary architectural search, we extensively tested larger vision encoders, including ViT-Base. However, empirical results indicated that employing larger visual backbones yielded only marginal gains in translation accuracy while incurring disproportionate parameter overhead and computational costs. By adopting ViT-Tiny, we ensure that our experiments isolate the effectiveness of the cross-modal fusion algorithms themselves, rather than relying on brute-force scaling of the visual encoder. The extracted 196 image patches from the ViT-Tiny are processed via our proposed Text-Guided Attention Pooling mechanism before interacting with the decoder.

### 3.3 Evaluation Metrics

Standard Machine Translation evaluations often fail to capture whether a model is genuinely utilizing multimodal information or simply exploiting language priors. To provide a comprehensive analysis, we employ two distinct evaluation frameworks:

**Linguistic Fluency (Test 1 & Test 2):** We use the standard BLEU metric [7] to measure general translation quality and structural fluency. Test 1 evaluates the in-domain performance on Multi30k, while Test 2 evaluates out-of-domain generalization on MSCOCO.

**Visual Disambiguation (Test 3):** To explicitly measure cross-modal grounding, we utilize the CoMMuTE benchmark. CoMMuTE is specifically designed as a binary forced-choice evaluation, requiring the model to resolve target-side ambiguities (e.g., gender, specific objects) using the provided image. Because it operates as a forced-choice between two options, a CoMMuTE score of approximately 50.00% represents statistical random chance. This metric serves as our primary indicator of modality collapse; any model scoring near the 50% threshold is fundamentally "blind" to the visual context, regardless of its corresponding BLEU score.

**Table 1.** Experimental results investigating the optimal depth of multimodal fusion and hybrid layer ratios. CoMMuTE scores highlight the persistence of modality collapse.

| Model Configuration       | Test 1 ( $\uparrow$ ) | Test 2 ( $\uparrow$ ) | Test 3 (% $\uparrow$ ) |
|---------------------------|-----------------------|-----------------------|------------------------|
| all-layers (Baseline DAD) | 40.95                 | 29.45                 | 50.22                  |
| bottom_heavy              | 41.29                 | 29.27                 | 49.78                  |
| top_heavy                 | 41.64                 | 29.35                 | 50.56                  |
| text1dual3 (Hybrid)       | 41.53                 | 29.61                 | 49.89                  |
| text3dual1 (Hybrid)       | 41.15                 | 29.82                 | 50.00                  |

## 4 Results

To empirically validate our hypotheses regarding structural depth and semantic alignment, we present our findings across three evaluation metrics: Test 1 (in-domain BLEU on Multi30k), Test 2 (out-of-domain BLEU on MSCOCO), and Test 3 (CoMMuTE visual disambiguation accuracy).

### 4.1 The Depth Illusion: Structural Search and Hybrid Ratios

We first evaluate the impact of altering the architectural positioning of the multimodal cross-attention layers. Table 1 summarizes the performance of various depth configurations and our extended hybrid ratio experiments.

The results yield a striking architectural paradox. In terms of linguistic fluency, the `top_heavy` configuration outperforms both `bottom_heavy` and the all-layers baseline on Test 1 (41.64 BLEU), supporting our initial hypothesis that visual features are best integrated at higher, more semantic abstraction levels. Building upon this, our hybrid ratio variants (`text1dual3` and `text3dual1`) demonstrated minor trade-offs between in-domain and out-of-domain fluency.

However, examining Test 3 exposes a critical failure: regardless of the fusion depth or the specific hybrid ratio of text-to-multimodal layers, the CoMMuTE accuracy remains stubbornly anchored at approximately 50.00%. Because CoMMuTE is a binary forced-choice metric, a 50% score is the statistical equivalent of random guessing. This proves that altering the structural depth of fusion is a "depth illusion"—it may optimize general language modeling, but it fundamentally fails to induce true visual grounding.

### 4.2 Impact of Semantic Alignment and Pooling

Having established that structural positioning alone cannot overcome modality collapse, we evaluate our semantic alignment techniques. Table 2 compares the baseline against different pooling strategies (CLS token) and our proposed Text-Guided Attention Pooling paired with the InfoNCE auxiliary loss.

The integration of InfoNCE alignment paired with our top-heavy configuration achieves the absolute highest in-domain translation fluency in our study, scoring 41.69 on Test 1. This confirms that mathematically synchronizing the

**Table 2.** Comparison of pooling strategies and InfoNCE alignment against the baseline models.

| Method             | Fusion Depth | Test 1 ( $\uparrow$ ) | Test 2 ( $\uparrow$ ) | Test 3 (% $\uparrow$ ) |
|--------------------|--------------|-----------------------|-----------------------|------------------------|
| baseline (DAD)     | all          | 40.95                 | 29.45                 | 50.22                  |
| baseline           | top_heavy    | 41.64                 | 29.35                 | 50.56                  |
| CLS pooling        | all          | 41.40                 | 29.93                 | 50.00                  |
| infoNCE            | all          | 41.56                 | 29.42                 | 50.00                  |
| infoNCE (Proposed) | top_heavy    | 41.69                 | 29.64                 | 49.78                  |

visual and textual semantic spaces prior to decoding prevents the introduction of disruptive noise, slightly enhancing general generation quality.

Yet, despite providing the decoder with perfectly aligned, text-guided visual features, the Test 3 CoMMuTE scores flatline at the random-chance threshold of  $49.78\% \approx 50.00\%$ . The model remains functionally blind to visual context when disambiguation is explicitly required.

## 5 Discussion and Failure Analysis

Our experiments yield a highly counterintuitive set of results: we successfully engineered methods to perfectly align text and vision without degrading—and even slightly improving—general BLEU scores, yet the core problem of modality collapse remains entirely unsolved. This forces a critical reevaluation of how multimodal transformers optimize their objectives.

### 5.1 The Limits of Encoder-Level Alignment

A primary takeaway from this study is the decoupling of encoder alignment and decoder utilization. Our InfoNCE auxiliary loss successfully forced the pooled visual vector to occupy the same semantic latent space as the text. However, providing a high-fidelity, aligned signal is functionally useless if the decoder refuses to process it. The persistent  $\sim 50\%$  CoMMuTE scores unequivocally demonstrate that encoder-level interventions are insufficient. Modality collapse is not a problem of misunderstood visual features; it is a problem of ignored visual features.

### 5.2 The Trap of Rigid Fusion and Language Priors

The root cause of this ignored signal lies within the mathematical rigidity of the decoder itself. The DAD architecture employs a static `0.5_sum` fusion mechanism. While we established this as the most stable baseline for BLEU scores, this stability is exactly what causes the collapse.

Neural networks are inherently lazy optimizers; they seek the path of least resistance to minimize cross-entropy loss. Because the Multi30k and MSCOCO datasets possess incredibly strong language priors—meaning the text alone is usually sufficient to guess the next word—the decoder learns that the mandatory

50% visual input is a liability. To prevent this "unnecessary" visual data from introducing noise during simple text-only generation, the model optimizes its attention weights to effectively zero out the visual stream.

Therefore, modality collapse is not an architectural bug, but a highly successful optimization strategy learned by the decoder to survive rigid fusion mechanics. This proves that as long as the fusion mechanism lacks the dynamic ability to mathematically penalize text-only reliance, the model will always default to its language prior, rendering even the most sophisticated visual alignment strategies obsolete.

## 6 Conclusion and Future Work

### 6.1 Conclusion

Our comprehensive investigation into multimodal fusion reveals a critical insight for the MMT community: optimizing the architectural depth of visual integration is entirely secondary to the foundational challenge of overcoming modality collapse. While our proposed Text-Guided Attention Pooling and InfoNCE alignment successfully synchronized the cross-modal semantic spaces and slightly improved general linguistic fluency, they fundamentally failed to induce true visual grounding.

The persistent  $\sim 50\%$  accuracy on the CoMMuTE benchmark explicitly proves that providing high-fidelity, aligned visual features to the decoder is insufficient. Because the rigid static fusion mechanism naturally favors the dataset's strong language prior, the model safely ignores the visual stream to minimize overall loss. Ultimately, we conclude that visual features, regardless of their architectural integration point or encoder-level quality, will continue to be ignored unless the decoder is mathematically compelled to utilize them.

### 6.2 Future Directions

To effectively break this strong language prior, our future work will pivot from structural routing to mathematical enforcement and fine-grained extraction.

**Multi-Query Attention Pooling:** We propose an evolution of our current text-guided pooling mechanism. Instead of compressing the entire image into a single global representation based on the whole sentence, we will extract specific visual vectors for individual linguistic elements (e.g., distinct nouns, colors, or subjects). Creating multiple highly specialized visual queries will allow the decoder to focus on fine-grained visual details directly relevant to specific target words, rather than attempting to decode a heavily compressed global vector.

**Bias-Mitigating Loss Functions:** More crucially, to combat the decoder's lazy optimization, our next step involves mathematical penalties. We plan to adapt bias-mitigating loss functions that actively penalize the model if it achieves a correct translation solely by relying on the textual context, without incorporating the visual input. This approach aims to mathematically enforce true multimodal dependency during training, fostering a mandatory reliance on visual data without compromising or artificially discarding valuable source information.

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